

TECHBIG SOLUTIONS

Ambattur, Chennai

Python Programming – Complete E2E Training Syllabus

Core Python | Automation | Web | Data Analytics | AI/ML | DevOps

Program Objective: Build complete practical Python skills from programming fundamentals through OOP, APIs, databases, automation, data analytics, machine learning, web development, testing, security, performance, deployment and real-world project implementation.

Reference basis: Python documentation and widely used Python ecosystem practices covering the language, standard library, packaging, testing, web development, data analytics, automation and deployment.

Program	Python Programming – Complete E2E
Level	Beginner → Advanced Developer / Data Analyst / Automation / AI
Core Areas	Python OOP SQL APIs Pandas NumPy Web Automation
Advanced Areas	ML/AI Data Engineering DevOps Security Testing Deployment
Suggested Duration	120–180 hours, customizable by batch and project depth

1. Python Programming Overview

- Python history, philosophy and use cases
- Python implementations and versions
- Interpreter, interactive shell and scripts
- Python ecosystem and package management
- Applications: web, automation, data, AI, testing and DevOps

2. Environment Setup & Development Tools

- Python installation and PATH
- IDLE, VS Code, PyCharm and Jupyter
- Virtual environments and pip
- requirements.txt and dependency management
- Project folders and interpreter selection
- Command-line execution and debugging

3. Python Syntax & Fundamentals

- Indentation, comments and statements
- Variables and naming conventions
- Dynamic typing and type checking
- Input/output
- Operators and expressions
- Type conversion and basic built-in functions

4. Data Types

- Numbers: int, float and complex
- Boolean and None
- Strings

- Lists, tuples, sets and dictionaries
- Mutable vs immutable objects
- Sequence operations and membership
- Type annotations overview

5. Strings & Text Processing

- Indexing and slicing
- String methods
- f-strings and formatting
- Unicode and encoding
- Regular expressions
- Text parsing and validation
- Practical text-processing programs

6. Conditional Statements & Loops

- if/elif/else
- Nested conditions
- for and while loops
- range and enumerate
- break, continue and pass
- Loop patterns
- Nested loops and practical exercises

7. Functions

- Defining and calling functions
- Parameters and return values
- Positional, keyword, default and variable arguments
- *args and **kwargs
- Scope and namespaces
- Lambda functions
- Recursion
- Documentation and clean function design

8. Collections & Data Structures

- List operations and comprehensions
- Dictionary operations and comprehensions
- Set operations
- Tuple unpacking
- Nested data structures
- Stacks, queues and basic data-structure patterns
- Choosing the right data structure

9. Object-Oriented Programming

- Classes and objects
- Constructors and instance attributes

- Methods and class methods
- Inheritance
- Encapsulation
- Polymorphism
- Abstract base classes
- Magic/dunder methods
- Composition vs inheritance

10. Modules, Packages & Imports

- import and from-import
- `__name__ == '__main__'`
- Creating modules and packages
- `__init__.py` concepts
- Absolute and relative imports
- Package structure
- Third-party package installation

11. Exception Handling

- Exceptions and error types
- try/except/else/finally
- Raising exceptions
- Custom exceptions
- Exception chaining
- Defensive programming
- Logging errors and troubleshooting

12. File Handling & Data Serialization

- Text and binary files
- with/open and file modes
- CSV and JSON
- Pickle concepts and security considerations
- Pathlib and filesystem operations
- File validation and batch processing

13. Regular Expressions

- Regex syntax and character classes
- Groups and quantifiers
- Search, match, findall and substitution
- Validation patterns
- Parsing logs and semi-structured text
- Practical regex exercises

14. Iterators, Generators & Functional Programming

- Iterable vs iterator
- iter and next

- Generator functions and yield
- Generator expressions
- map, filter and reduce
- zip and itertools
- Lazy evaluation and memory-efficient processing

15. Decorators & Context Managers

- First-class functions
- Closures
- Decorators
- Parameterized decorators
- functools
- Context managers
- Creating custom context managers
- Practical reusable patterns

16. Type Hints & Modern Python

- Variable and function annotations
- typing module
- Generic collections
- Optional/Union concepts
- Protocols overview
- Dataclasses
- Modern Python syntax and best practices

17. Testing & Code Quality

- Unit testing with unittest
- pytest concepts
- Fixtures and parametrization
- Mocking
- Assertions and test organization
- Coverage concepts
- PEP 8 and linting
- Formatting and code-review practices

18. Logging & Application Diagnostics

- logging module
- Log levels and handlers
- Formatters
- File and console logging
- Structured logging concepts
- Debugging workflows
- Production troubleshooting

19. SQL & Database Programming

- Relational database concepts
- SQL fundamentals
- SQLite with Python
- Parameterized queries
- Transactions and connection management
- PostgreSQL/MySQL connectivity concepts
- ORM overview and database best practices

20. Web Development with Python

- HTTP and web fundamentals
- Flask fundamentals
- FastAPI fundamentals
- Routing and request handling
- Templates and APIs
- JSON REST APIs
- Validation and error handling
- Application configuration

21. REST APIs & Integration

- HTTP methods and status codes
- REST architecture
- requests/httpx concepts
- Authentication: API keys, OAuth concepts and tokens
- JSON request/response handling
- Pagination and retries
- API error handling
- Building and consuming REST APIs

22. Web Scraping & Automation

- HTML structure and parsing
- Requests and BeautifulSoup concepts
- Browser automation overview
- Data extraction and cleaning
- Scheduling and batch automation
- Robots.txt, terms of service and responsible scraping
- Automation error handling

23. Data Analysis with NumPy

- Arrays and vectorized computation
- Array indexing and slicing
- Broadcasting
- Reshaping and aggregation
- Mathematical operations
- Missing-value concepts

- NumPy performance patterns

24. Data Analysis with Pandas

- Series and DataFrame
- Reading CSV/Excel/JSON/SQL data
- Selection, filtering and sorting
- GroupBy and aggregation
- Merge, join and concat
- Pivot tables and reshaping
- Missing data and duplicates
- Date/time analysis
- Performance and practical analytics

25. Data Visualization

- Matplotlib fundamentals
- Line, bar, scatter and histogram charts
- Subplots and annotations
- Pandas plotting
- Seaborn concepts
- Interactive visualization overview
- Dashboard-ready visualization principles

26. Machine Learning with Python

- ML workflow and problem types
- Feature engineering
- Train/validation/test split
- Regression
- Classification
- Clustering
- Preprocessing and pipelines
- Model evaluation and cross-validation
- scikit-learn ecosystem

27. Artificial Intelligence & Deep Learning Overview

- AI/ML/DL relationship
- Neural-network fundamentals
- Tensor concepts
- PyTorch/TensorFlow overview
- Model training concepts
- Embeddings and transformers overview
- Generative AI and LLM application concepts

28. Python for Data Engineering

- ETL/ELT concepts
- Batch processing

- Data pipelines
- CSV/JSON/API ingestion
- Database loading
- Data validation
- Scheduling concepts
- Airflow overview
- Cloud data-engineering concepts

29. Python for DevOps & Automation

- System administration scripting
- Environment variables
- Process and subprocess management
- OS and filesystem automation
- Configuration management
- CI/CD scripting concepts
- Docker integration overview
- Cloud SDK/API automation

30. Security & Secure Python Development

- Secrets and credential management
- Environment variables and secret stores
- Input validation
- SQL injection prevention
- Safe file handling
- Dependency security
- Secure API practices
- Authentication/authorization concepts
- Common Python security pitfalls

31. Performance & Concurrency

- Time and space complexity basics
- Profiling
- Optimization strategies
- Multithreading
- Multiprocessing
- asyncio and asynchronous programming
- Concurrent API calls
- Memory-efficient processing

32. Packaging & Deployment

- Project packaging concepts
- pyproject.toml overview
- Dependencies and lockfiles
- Build and distribution concepts

- Application configuration
- Docker deployment
- Cloud deployment overview
- Production environment practices

33. Git, GitHub & Team Development

- Git fundamentals
- Branches and merging
- Pull requests
- Conflict resolution
- .gitignore
- Repository structure
- Code review and collaboration
- CI concepts

34. Real-Time E2E Projects

- Employee/HR management automation system
- Inventory management application with database
- REST API using FastAPI with authentication
- Web application using Flask
- Excel/CSV automation and reporting solution
- Sales analytics using Pandas, SQL and visualization
- ETL pipeline: API → Python → database → reporting
- Web scraping and scheduled data pipeline
- Machine-learning prediction application
- End-to-end production Python application with testing, Docker and CI/CD

35. Troubleshooting & Production Support

- Environment and dependency issues
- Import/package failures
- Runtime and exception analysis
- Database connectivity issues
- API timeout and authentication failures
- Performance problems
- Logging and monitoring
- Deployment failures
- Root-cause analysis

36. Interview & Certification Preparation

- Python programming interview questions
- Data-structure and algorithm exercises
- OOP and design questions
- SQL and Python integration
- Pandas/NumPy practical questions

- API and web-development scenarios
- Testing and debugging questions
- Real-time project explanation
- Coding assessment and mock interview

Recommended Hands-on Deliverables

- Build a Python application using OOP, file handling, exceptions, logging, testing and a database.
- Develop and consume a production-style REST API with authentication, validation, error handling and documentation.
- Create a complete data-analysis project using SQL, Pandas, NumPy and visualization.
- Develop an automated ETL pipeline from APIs/files into a database with validation and scheduled execution.
- Build a machine-learning project with preprocessing, training, evaluation and model-serving concepts.
- Containerize and deploy a Python application with Git, testing and CI/CD practices.

Training Outcome

Learners completing this program should be able to develop maintainable Python applications, automate business processes, work with databases and APIs, perform data analysis, build introductory AI/ML solutions, write tests, troubleshoot production systems and deploy Python applications using modern development practices.

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